Fermat vs Descartes Paper Outline

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Topic: Analyzing the Feud between Fermat and Descartes over discovery of a tangent line of a curve.

(Potential Titles: The Legacy of the Descartes-Fermat Feud on Analytic Geometry and Calculus, Analyzing the Methods of Descartes and Fermat for Finding Tangent Lines)

I. **Introduction:** The focus of this paper will be about how Descartes and Fermat seemingly came up with a method to compute the tangent line of a curve at any specific point without the usage of calculus. The mathematicians did not see eye to eye on how the other came to their respective conclusions, but as a result, this feud helped form the basic intuitions of calculus that would be picked up by Newton and Liebniz for the foundations of Calculus.

II. **Body:**

A. **Key Point 1:** This section will give background information on the origins of the Tangent Line and explain the Tangent Line Problem and why it is important.

1. Early developments of the tangent line come from early Greek ideas of analytic geometry. Specifically mathematicians Euclid (300 BC) and Apollonius (225 BC) had similar definitions of tangent lines. This is known by their individual works, *Elements (Euclid, Book 3 [Fundamentals of Plane Geometry Involving Circles] Proposition 18)* and *Conics (Apollonius, Book 1 [Definitions] Proposition 17).* Although their works spoke of tangents to circles, these works paved the way to many mathematical ideas going forward and would be picked up by Fermat and Descartes in the 17th Century.

   Sources:

   Fitzpatrick, Richard. EUCLID'S ELEMENTS of GEOMETRY. 2007 English Translation By Fitzpatrick.

   Rosenfeld, Boris. APOLLONIUS of PERGA CONICS. BOOKS ONE - SEVEN English Translation By Rosenfeld.

2. The tangent line problem involves finding the slope of a curve at a specific point. This problem was studied by Fermat in the 17th century, who developed a method for finding the slope of the tangent line using the limit of the difference quotient. Simultaneously, Descartes formalized ideas that related geometric shapes and algebraic techniques with the development of the Cartesian coordinate system in his work, *La Géométrie,* which united algebra and geometry into a unified subject named Analytic

Geometry. With this unification came Decartes' own method for calculating the tangent line with normal lines, circles and algebraic methods.

Sources:
La Géométrie By Rene Descartes (Translated by David Eugene Smith and Marcia L. Latham)

3. The tangent line problem is important in a variety of fields, including physics, engineering, and economics. For example, in physics, the slope of the tangent line of a displacement graph represents the velocity of an object at a specific point in time. It also extends to higher order principles such as partial derivatives.

B. **Key Point 2:** This Section will Explain Rene Descartes' Method of computing the tangent line of a curve with the Method of Normals, limitations of the method.
   1. Rene Descartes developed a method for finding the tangent line to a curve at a given point, known as the method of normals. The method involves finding the line perpendicular to the tangent line at the point of interest, called the normal line. This tangent equation would also be a tangent to some arbitrary circle with radius with length of the original point to the x-axis in a straight line. Then with algebraic manipulations of these found values, eventually the slope of the normal line was found which happened to be the negative reciprocal of the original line. (Perhaps here a diagram would be more effective in showing this)

      Sources:
      La Géométrie By Rene Descartes (Translated by David Eugene Smith and Marcia L. Latham)

   2. This method was plausible but was very complex for curves represented by polynomials. Also this method required the curve in question to be explicitly defined. In other words, implicitly defined functions were seemingly not possible to have a method for finding a tangent line at a given point

      Sources:
      La Géométrie By Rene Descartes (Translated by David Eugene Smith and Marcia L. Latham)

C. **Key Point 3:** This section will explain Pierre De Fermat's Method of computing the tangent line of a curve via Adequality.
    1. Pierre de Fermat developed a method for finding the tangent line to a curve using the concept of adequality, which is a notion of equality between infinitesimal quantities. Fermat's approach involves comparing two infinitesimal quantities associated with the curve at the point of interest, one corresponding to a small change in the value of x and the other corresponding to the corresponding change in the value of y. By setting these two infinitesimal quantities equal to each other, Fermat was able to find the slope of the tangent line at the point of interest.

        Sources:
        Methodus ad disquirendam maximam et minimam et de tangentibus linearum curvarum by Pierre De Fermat (Communicated by C. B. Boyer)

    2. Whenever publishing works, Fermat was infamous for leaving some steps in his proofs ambiguous which left the professional mathematics community to lean more towards Descartes' method initially, as it was deemed to be more rigorous than Fermat's. One example of Fermat's informal style is his so-called "Fermat's Last Theorem," which he claimed to have proven but for which he never wrote down a proof that was widely accepted by the mathematical community. Fermat's proof was reportedly written in the margin of a book and was never published during his lifetime. It was not until centuries later, when mathematicians Andrew Wiles and Richard Taylor independently proved the theorem using modern mathematical techniques, that Fermat's claim was finally confirmed.

        Sources:
        "Pierre de Fermat" by Andrew Granville, in The Princeton Companion to Mathematics edited by Timothy Gowers, June Barrow-Green, and Imre Leader

D. **Key Point 4:** This section will go over the rivalry the two mathematicians went through relating specifically to the tangent line and how their legacy eventually led to the discovery of calculus.

    1. The rivalry between René Descartes and Pierre de Fermat was long-standing and often believed to be a contentious one. The two mathematicians had differing ideas about the nature of mathematical

inquiry, and their disagreements over the tangent line to a curve were just one manifestation of this broader divide.

Sources:
"Mathematical Giants: Rene Descartes and Pierre de Fermat" by Peter Rowlett, Mathematics Today
"Descartes and Fermat: Early Pioneers of Calculus" by David D. Nolte, American Journal of Physics

2. This feud was then turned into a challenge from Descartes to Fermat when Descartes challenged Fermat to find the tangent line of a specific curve. The famous curve, Folium of Descartes, was proposed to have no way to calculate the tangent line at any given point by Descartes as it was implicitly defined: $x^3 + y^3 + 3dxy$, where d is simply a constant. The goal of this challenge was to show Fermat's method was inferior to his own. Descartes was proven wrong by Fermat and as a result, Fermat was given more notoriety for works.

   Sources:
   Calculus Gems By G. F. Simmons (Page 101)

3. Given the nature of Fermat's method of adequality involving infinitesimally small quantities and the hindsight of present day Calculus, it can be assumed this way of thinking led to the discovery of Calculus. Additionally, the feud between Descartes and Fermat helped to popularize the study of analytic geometry and paved the way for the development of the calculus of variations, a field of mathematics concerned with finding optimal solutions to problems involving functional relationships.

   Sources:
   The History of Calculus and Its Conceptual Development by Carl B. Boyer and Uta C. Merzbach

III. **Conclusion:**

In conclusion, the feud between René Descartes and Pierre de Fermat over their respective methods for finding the tangent line to a curve highlights the important role of mathematical controversy and debate in the development of mathematics. While both mathematicians made significant contributions to the field of calculus, they approached the problem of finding the tangent line in different ways, with

Descartes favoring the method of normals and Fermat relying on adequality and similar triangles.

Ultimately, the feud between Descartes and Fermat serves as a reminder of the importance of debate and disagreement in the pursuit of mathematical truth, and the vital role that collaboration and competition play in advancing our understanding of the world around us.

IV. **References:**

[1] Boyer, Carl B., and Uta C. Merzbach. The History of Calculus and Its Conceptual Development. Dover Publications, 2014.

[2] Descartes, Rene. La Géométrie. Translated by David Eugene Smith and Marcia L. Latham, Dover Publications, 1954.

[3] Fitzpatrick, Richard. Euclid's Elements of Geometry. 2007 English Translation by Fitzpatrick, Lulu Press, Inc., 2007.

[4] Granville, Andrew. "Pierre de Fermat." The Princeton Companion to Mathematics, edited by Timothy Gowers, June Barrow-Green, and Imre Leader, Princeton University Press, 2008, pp. 251-255.

[5] Rosenfeld, Boris. Apollonius of Perga Conics. Books One - Seven English Translations by Rosenfeld, Springer, 2016.

[6] Simmons, G. F. Calculus Gems. McGraw-Hill, 1992.